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Development of An Electrostatic Actuator Exceeding 10N Propulsive Force

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ABSTRACT

This paper presents a new type of electrostatic actuator. It is suited for small scale applications as well as for the macroscopic world up to now dominated by electromagnetic devices. A propulsive force of 21N has been achieved with a ±1000V excitation voltage.

The new actuator basically consists of a stack of ten thin actuators developed by the authors. The stacked actuator is 120mm×110mm×5mm in overall size and 70g in weight. It is equipped with a position sensor with an 80μm resolution and embedded in a closed-loop control improving speed and power performance by a factor of about two, reaching top values of 16cm/sec and 1.2W of mechanical power. The maximum values of force/weight and power/weight ratio of the stacked actuator are 290N/kg and 17W/kg respectively.

INTRODUCTION

At the present time, large scale motion world is dominated by electromagnetic actuators. However, since their power/weight or power/volume ratio is not satisfactory, mechanical systems which employ electromagnetic actuators are heavy and occupy large space. Besides conventional electromagnetic actuators, various new actuators such as pneumatic muscles [1,2], actuators using shape memory alloy [3], mechanochemical actuators using gels [4], ultrasonic actuators [5,6] etc have been studied. Electrostatic actuators have also been developed [7,8]. However none of them can produce forces comparable to those of conventional actuators at the present time.

In the small scale world, electrostatic actuators are more advantageous than electromagnetic actuators, and many electrostatic actuators with high performance have been reported [8,9,10,11]. As the size of an electrostatic actuator is reduced, its force/volume ratio is increased. Therefore, a very small electrostatic actuator can exceed other ordinarily sized actuators in force/volume ratio.

By connecting many very small electrostatic actuators with large force/volume ratio, we can get an ordinarily sized actuator which has a larger force/volume ratio than conventional actuators. However, it is difficult to connect the formerly developed actuator. In order to make the most of the size effect of the electrostatic actuator, we must design them in view of their use as modular components of an stacked actuator.

We have formerly developed a *film actuator* [12] which consists of a stator film having electrodes and a slider made of slightly conductive material. Since the slider of the *film actuator* do not have electrodes, the slider and the stator do not have to be positioned accurately. However, its operation is not stable since resistance of the slider film is affected by the environment of the actuator. In order to overcome this problem, we developed a variable capacitance motor with active slider (VCMA). Since both the slider and the stator of the VCMA have electrodes, they have to be aligned accurately. As a prototype, ten VCMAs were stacked up to an actuator with ordinary size, which was named stacked VCMA (SVCMA). Operation of the SVCMA and its force/weight ratio are more stable and larger than those of stacked *film actuator*. This paper describes the principle, analysis and closed-loop control of a VCMA, and the fabrication and test of an SVCMA.

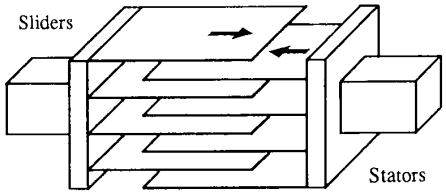


Figure 1 Schematic Figure of SVCMA. An SVCMA consists of many VCMAs stacked up.

CONNECTED ELECTROSTATIC ACTUATOR

The size effect on an electrostatic actuator made up of smaller units is described in comparison with the effect on a Permanent Magnet (PM) electromagnetic actuator. Essential points in development of a modular actuator are then discussed.

The effect on an actuator of varying both the size as a whole and that of each unit can be described as

$$F \propto R^n r^m \qquad (1)$$

where F is the force produced by the actuator, R is the characteristic dimension of the modular actuator, r is the size of its unit, n and m are numbers dependent on the actuator.

The force generation F_m by a PM electromagnetic is estimated as (See APPENDIX)

$$F_m \propto R^{2.5} r^0 \qquad (2)$$

This shows that there is neither merit nor demerit in dividing an electromagnetic actuator into small units and re-assembling them. On the other hand, The force of an electrostatic actuator F_s is estimated as

$$F_s \propto R^3 r^{-1} \qquad (3)$$

This says that the smaller size each unit of an electrostatic actuator has, the larger total force it produces. Therefore, if the size of the unit is small enough, a modular electrostatic actuator as a whole can generate a larger force than an electromagnetic actuator.

To make the most of the scale effect of the electrostatic actuator, its module must have the properties shown below.

1. In order to obtain a large total output, a large number of modules is needed. Therefore, the module must be mass-producible.
2. Generally speaking, connecting a lot of small things in right order is not easy. We should design each module so that its structure makes the connection easier.
3. In order to obtain large total output, we must accumulate the outputs with small loss. If the outputs are gathered through an energy wasting transmission mechanisms, the outputs of the units are wasted

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in the mechanism. Therefore, the modules must have a structure where their outputs can be connected directly.

STRUCTURE, PRINCIPLE AND MERITS OF VCMA

Structure The SVCMA is a stacked VCMA (Figure 1). Each VCMA consists of two elements, a thin stator and a slider (Figure 2). Electrodes are fabricated both on the stator and the slider. Those of the stator and the slider have 3-phase and 4-phase structure, and their pitches are in the ratio of 4:3. These seven phases of the electrodes can be excited independently. The primitive unit of VCMA consists of three stator electrodes and four slider electrodes. The symbol u in Figure 3 is the length of the unit.

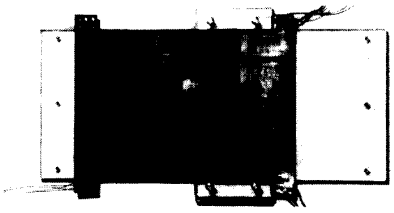


Figure 2 Photo for a VCMA

Operating Principle

The operation of the VCMA is as follows.

(a) When an excitation shown in Figure 3 (a) induces charges of equal polarity into the facing electrodes, repulsive force is generated between the slider and the stator. The normal components of the force levitates the slider or reduces the friction between the stator and the slider, and the tangential component drives the slider forward.

(b) As the slider moves, the distance between charges in the electrodes grows larger. Since the distance reduces the force caused between the stator and the slider, the slider stops moving shown in Figure 3 (b).

(c) The excitation pattern is switched in rotation as depicted in Figure 3 (c). Relative position of the charges is similar to those of Figure 3 (a). The slider starts moving again. By this operation, all the charges of the stator and sliders are moved rightward respectively by a pitch of the electrodes where they induced; pitches of the electrodes in the stator and the slider are $u/3$ and $u/4$ respectively. The switching operation reduces the distance between charges in the slider and the stator by $u/12$.

As these procedures are repeated, the slider moves on.

If the displacement in procedure (a) is much larger than $u/12$, the single operation in procedure (c) can not move the slider again. However, the repeated operations will decrease the distance enough to regain the repulsive force to move the slider again at last.

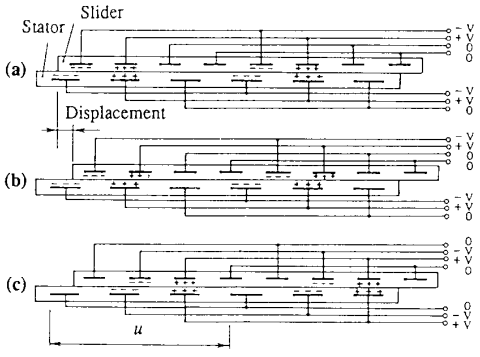


Figure 3 Principle of SVCA. Switching excitation voltage in rotation, the slider is driven.

Merits

VCMA is designed so that it is well suited as a module of SVCMA. The merits of the structure and the operation of the VCMA as a module of a modular electrostatic actuator are as follows.

(a) Since the same polarity of charges are directly put into facing electrodes of the stator and of the slider, repulsive force can be generated between them. Since the repulsive force reduces the friction, the VCMA needs no bearing. It facilitates fabrication of a VCMA. The VCMA is unique in this point. Many actuators similar to VCMA have been reported, but most of them do not have multiple phases of electrodes which can be excited independently both in their stator and slider (rotor). An actuator whose electrodes have such a structure can not generate repulsive force.

(b) A pitch of electrodes is a characteristic dimension of the unit of an electrostatic actuator. In order to generate a large force, an electrostatic actuator with small pitches of the electrodes is needed. Since the electrodes of VCMA have a 2-dimensional structure with small pitches, they can be easily fabricated by photo lithography process.

(c) The VCMAs are easily stacked up to an SVCMA. This is an important advantage of the VCMA as a component of a modular actuator.

(d) The structure of the VCMA reduces the energy loss in the power gathering processes of the SVCMA. The force generated by the units of a VCMA are transmitted through its slider and stator as a form of tension and gathered at the ends of the two elements. No energy wasting mechanism is needed.

ANALYSIS

The relation of force and the slider's position is analyzed by the surface charge method (SCM).

Assumptions

Figure 4 illustrates a model of the VCMA. The primitive unit of the actuator consists of electrodes a_1, a_2, a_3, a_4, b_1, b_2 and b_3, which are repeated horizontally. For simple computation, we assume that: (1) the electrodes are long enough to use the two dimensional field approximation; (2) the thickness of the electrodes is negligible; (3) all the media have a uniform permittivity and (4) there is a ground plane $100u$ below the stator electrodes.

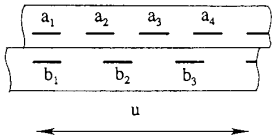


Figure 4 A model of operational unit of VCMA

Results

There are many possible excitation patterns which can operate the VCMA other than shown in Figure 3. Two voltage patterns shown in Table 1 are used in the computation. Permittivity is defined $\varepsilon = 3.5\varepsilon_0$, where ε_0 is that of vacuum; $3.5\varepsilon_0$ is the permittivity of polyimide which is used as substrate of the electrodes. Excitation voltage of ±1000V is adopted.

Table 1 Excitation voltage patterns.

	VP_1	VP_2
$a_1\ a_2\ a_3\ a_4$	+ − 0 0	+ − + +
$b_1\ b_2\ b_3$	+ − 0	+ − +

Figure 5 shows a calculation results of the relation of forces and the slider's displacement. The thick line and the thin line correspond to the driving force and the levitation force. Since these diagrams are repeated at each interval of u, the forces only inside one step u are displayed. Displacement is expressed in phase position ϕ, where ϕ is defined to complete a full cycle (360°) for $x=u$. The origin of the ϕ axis is moved to the phase where the driving force is 0.

When the slider stands still, the friction force between stator and slider is described as

$$f_F = -sgn(f_D)\cdot\mu(mg-f_L) \quad (4)$$

where f_F is the friction force, f_D is the driving force, μ is the friction coefficient, mg is weight of the slider, and f_L is the levitation force. Since the friction force reduces the horizontal force, the slider is driven by the net driving force f_N

$$f_N = f_D - f_F \quad (5)$$

f_N is displayed by the broken line in Figure 4 (b). When $f_N > 0$, slider starts in positive direction. When $f_N < 0$, it starts in the negative direction.

When the excitation pattern is rotated, the charges of the slider slides by 30° ($u/12$) against the charges of the stator. The regions where f_N is positive and where it is negative are wider than 30° respectively. Therefore, the slider can be started in both directions wherever it is located.

The results of the computation say that the pattern VP2 can produce larger f_D than VP1. We adopt the pattern VP2 when we test the VCMA and the SVCMA.

CLOSED-LOOP CONTROL

The operating speed of VCMA is increased as the switching period is decreased. However, at a certain period, the operation often goes out-of-step. A closed-loop control as follows reduces this effect and improves the operational speed of the VCMA.

By switching the excitation pattern in rotation, we can obtain twelve kinds of f_N, whose f_N-ϕ diagrams are different in phase by 30°. Closed-loop control observes the phase of the slider and switches the excitation so as to generate the largest f_N at the observed phase. In order to select the optimal f_N, the closed-loop control needs a position sensor with 30° resolution.

SENSOR

In order to apply the closed-loop control described above, we developed an electrostatic position sensor and equipped it with VCMA. The sensor employs electrodes with the same structure as those for actuation. Figure 6 shows its basic structure and principle.

The four phases in the slider are connected to high frequency of 100kHz which differ in phase by 90°, and the three phases in the slider are connected to the ground through current detectors. The high frequency applied to the slider electrodes cause an electric field below the slider, and the field induces currents into the stator electrodes. Each current is sinusoidal, and its phase is dependent on the position of the slider. Therefore, the three phases of currents I_1, I_2 and I_3 can be approximated as

$$\begin{aligned} I_1 &\approx A\cos(\omega t+\phi) \\ I_2 &\approx A\cos(\omega t+\frac{120}{180}\pi+\phi) \\ I_3 &\approx A\cos(\omega t+\frac{240}{180}\pi+\phi) \end{aligned} \quad (6)$$

where ω is the angular frequency of the high frequency voltages, the ϕ is displacement in normalized phase position. These currents are detected by three chopper detectors. Their outputs, V_1, V_2 and V_3 are

$$\begin{aligned} V_1 &= \frac{1}{T}\int_0^T \sin(\omega t)\cdot I_1(t)dt = k\sin(\phi) \\ V_2 &= \frac{1}{T}\int_0^T \sin(\omega t)\cdot I_2(t) = k\sin(\phi+\frac{2}{3}\pi) \\ V_3 &= \frac{1}{T}\int_0^T \sin(\omega t)\cdot I_3(t) = k\sin(\phi+\frac{4}{3}\pi) \\ k &= -\frac{1}{2}A \end{aligned} \quad (7)$$

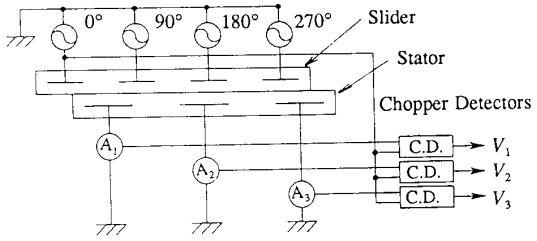


Figure 6 Principle of sensor. The outputs of the chopper detectors are encoded by PC.

Though ϕ can be calculated in an analogue value from these outputs, the closed-loop control needs only 30° resolution. Then we use only the sign of the detectors' outputs. Since, zero cross points of V_1, V_2 and V_3 divide 360° into six equal parts, they can be graduation of a position sensor with 60° resolution. Since 30° is needed, define V_{12}, V_{23} and V_{31} as

$$V_{12} \equiv V_1 - V_2, \quad V_{23} \equiv V_2 - V_3, \quad V_{31} \equiv V_3 - V_1 \quad (8)$$

They are

$$\begin{aligned} V_{12} &= \sqrt{3}\,k\sin(\phi+\frac{5\pi}{3}) \\ V_{23} &= \sqrt{3}\,k\sin(\phi+\frac{\pi}{3}) \\ V_{31} &= \sqrt{3}\,k\sin(\phi+\pi) \end{aligned} \quad (9)$$

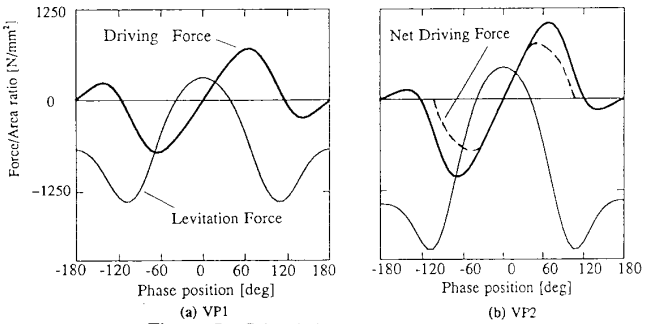


Figure 5 Caluculation result of the relation of forces and the sliders displacement. Forces is normalized by force/area.

Since zero cross points of V_1, V_2, V_3, V_{12}, V_{23} and V_{31} divides the 360° into twelve equal parts, we can obtain a desirable position sensor with 30° resolution.

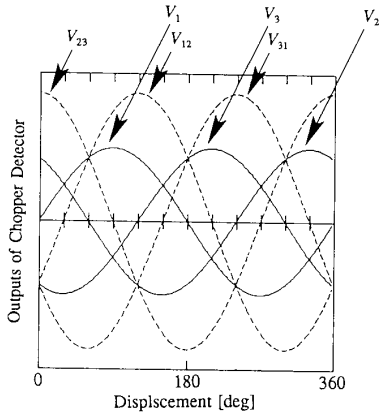


Figure 7 Outputs of the chopper detector and additional signals calculated from them. They divide the full cycle into equal twelve parts.

FABRICATION

VCMA

The electrodes for actuation and sensing are placed as shown in Figure 8. Two blocks of sensor electrodes are located on both sides of actuation electrodes so that error caused by yaw of slider will be reduced. On the stator, there are 312 lines of electrodes. On the slider, there are 416 lines. Common lines, which connect the same phases of electrodes, lead to the terminals on the tabs.

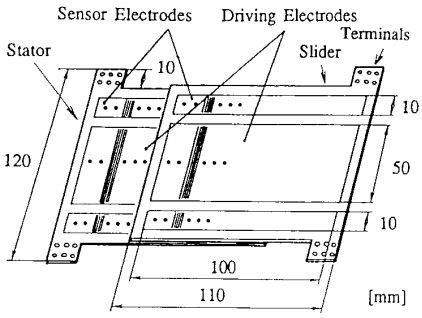


Figure 8 Location of the electrodes. Common lines are omitted for simplification.

The electrodes are fabricated on polyimide films. Both sides of them are covered with film of the same material. Figure 9 illustrates the cross sectional view of the VCMA. When a single VCMA is operated, dielectric fluid is injected between the stator and the slider. The weight of a VCMA including the fluid is approximately 7g.

Assembly

Ten sets of VCMAs are stacked as shown in Figure 10. Steel pins connect the VCMAs both mechanically and electrically. Spacers of anisotropicly conductive rubber are inserted between sliders and between stators so that they will not press themselves each other. It improves conductivity between terminals of the same phase in the different VCMA, and maintains isolation between different phases since it is isotropic. The actuator is immersed into dielectric fluid with $1.9\varepsilon_0$ permittivity. Net weight of SVCMA including the fluid is 70g.

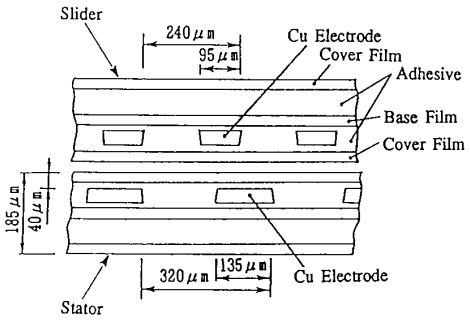


Figure 9 Cross sectional view of a VCMA. The slider and the stator is fabricated by the same process as that for flexible print circuit.

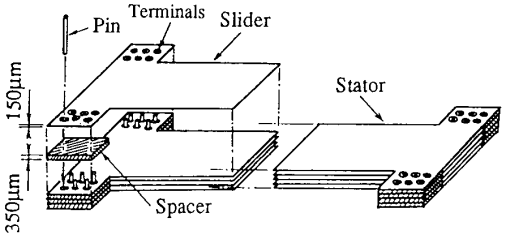


Figure 10 Assembly of SVCMA. The steel pins and the spacers connect the VCMAs electrically.

EXPERIMENTAL RESULTS

One set of VCMA

Figure 11 shows the stepwise movement of the slider of the VCMA operated by an open-loop control at excitation voltage ±800V. Figure 12 shows the relation of the produced force and excitation voltage. The force is proportional to the square of the voltage. The maximum value of that is dominated by the breakdown voltage between the stator electrodes and the slider electrodes. The dielectric fluid improves the maximum force, since it increase the breakdown voltage from 600V to 1000V. In addition, large permittivity of the fluid increases the force at the same voltage. The largest force generation with $1.9\varepsilon_0$ dielectric fluid injected is 1.3N at ±1000V. The maximum speed at ±800V is 9.6cm/sec.

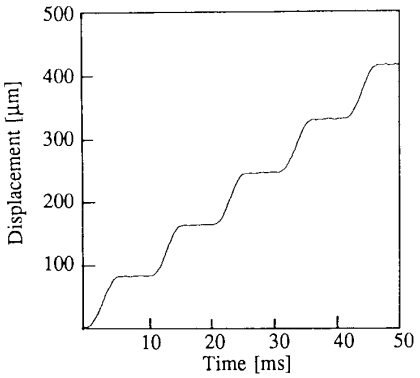


Figure 11 Stepwise movement of the slider of the VCMA observed by an optical motion sensor.

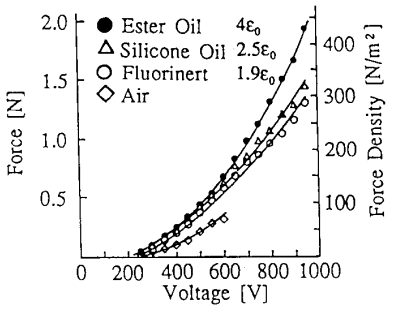


Figure 12 Force generation and excitation voltage of VCMA

SVCMA

Figure 13 shows the relation of force generation of SVCMA and excitation voltage. The output force is proportional to the square of the excitation voltage. The maximum force is 21N at ±1000V. Since interaction of multiple stator and slider increases the force generation of the SVCMA, the force of an SVCMA is larger than ten times the force produced by a single VCMA. The maximum speed at ±800V is 11cm/sec. The maximum power generation is 0.55W at 4.0cm/sec, 14N and ±1000V respectively in speed, force and excitation voltage.

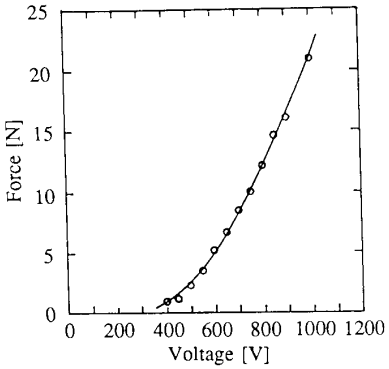


Figure 13 The Relation of force generation and excitation voltage of SVCMA

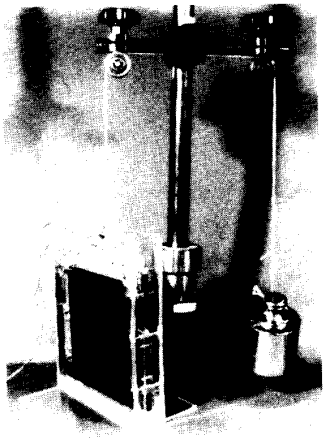


Figure 14 Measurement of force generation. SVCMA immersed in the dielectric fluid is pulling up a weight of 1kg.

Sensor and Closed-loop Control

Figure 15 shows the encoded outputs of the sensor. A resolution of 80μm, i.e. 30° in phase, is achieved. Since the effect of the yaw is not removed perfectly, there are small irregularities left.

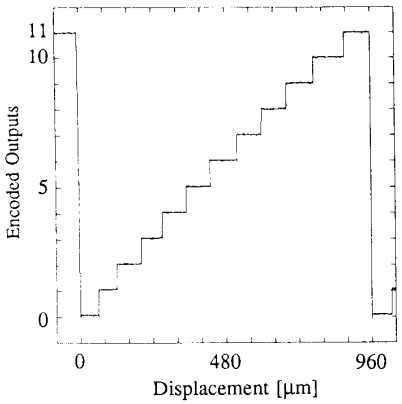


Figure 15 The outputs of the sensor. The output of the chopper detector is encoded to 12-step outputs by a PC. The steps divide 960μm, *u*, into twelve equal parts.

Figure 16 shows the performance of the closed-loop control operating a single VCMA. The slider traverses smoothly without missing any step. The maximum speed is increased to 40cm/sec. Figure 17 is the relation of the speed and the force generation of the SVCMA operated by the closed-loop control. The closed-loop control improves the velocity especially under small load. The maximum speed of SVCMA at ±800V is increased to 16cm/sec. The maximum power generation by the closed-loop control is 1.2W at 11cm/sec, 1.1N and ±1000V respectively in speed, force and excitation voltage.

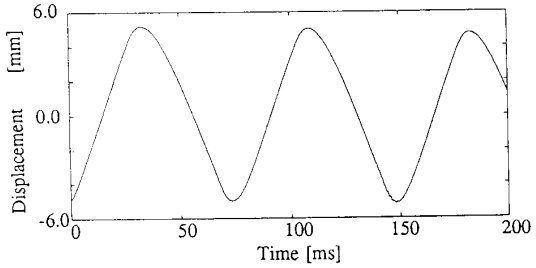


Figure 16 Two-way movement of slider of the VCMA operated by the closed-loop control.

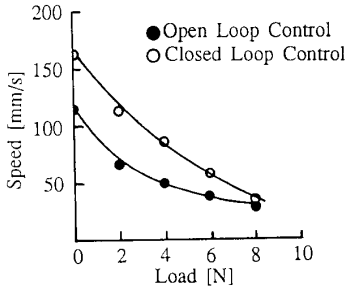


Figure 17 The relation of speed and force of VCMA. Maximum speed was measured under various loads.

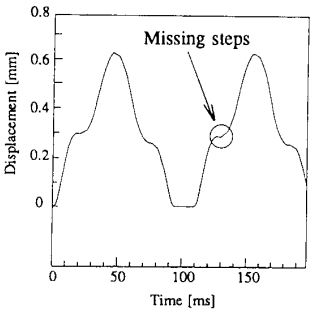


Figure 18 Movement of the slider of the VCMA operated by open-loop control. Out-of-steps are observed.

Table 2 Maximum performances of SVCMA. All the values other than speed are those at ±1000V. Speeds are those at ±800.

	Open-loop control	Closed-loop control
Force	21N	13N
Power	0.55W	1.2W
Speed*	11cm/sec	16cm/sec
Force/weight	290N/kg	180N/kg
Power/weight	7.7W/kg	17W/kg

CONCLUSION

We have proposed a *variable capacitance motor with active slider*, VCMA. Ten VCMAs are stacked, to be a *stacked variable capacitance motor with active slider*, VCMA. The prototype of SVCMA can produce 21N. Its force/weight ratio and power/weight ratio are comparable to those of conventional electromagnetic actuators. By reducing the pitch of the electrodes, the excitation voltage ±1000V can be lowed for more practicality. Since most of the supplied energy is lost in the driver circuit of SVCMA, the efficiency of the SVCMA including the circuit is approximately 5%. We are planning to improve the this value with LC resonator.

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APPEDIX

Size Effect of A Permanent Magnet Electromagnetic Actuator

Let characteristic dimension of the actuator and that of its units be R and r.

According to the Lorentz's law, force generation of the single unit of electromagnetic actuator is

$$f_m \approx B \cdot r^2 j \cdot r$$

where B is field of the permanent magnet, j is the current density of the winding. The heat generation of a unit is

$$q \approx \rho \cdot j^2 \cdot r^3 \quad (11)$$

Assume that the total heat radiation Q of the actuator is dominated by the surface area of the actuator.

$$Q \propto R^2 \quad (12)$$

where ρ is the moment of resistance of the winding. Q is balanced with the total heat generation.

$$q \propto \frac{r^3}{R^3} \cdot Q \approx \frac{r^3}{R}$$

Therefore, the total output is

$$F_m \propto \frac{R^3}{r^3} q^{0.5} r^{-1.5} = R^{2.5} r^0 \quad (14)$$

Size Effect of An Electrostatic Actuator

The maximum force generation of electrostatic actuator can be estimated with the maximum value of Maxwell's stress. It is given by breakdown field. Force generation of the unit is

$$f_s \approx E^2 r^2 \quad (15)$$

where R and r correspond to those for an electromagnetic actuator as defined above. Therefore total force is

$$F_s = \frac{R^3}{r^3} f_s \propto R^3 r^{-1} \quad (16)$$